

Remote Sensing Analysis Report

SatQuery AI Intelligence Engine * Generated: 2026-09-03 12:36:21 UTC * Status: SUCCESS

ANALYSIS ID
SQ-44A8ADB527

DETECTED TASK
Text-Guided Region Grounding

INPUT TYPE
Single Satellite Scene

SENSOR FAMILY
Sentinel-2 MSI

MODALITY
Optical

SESSION / THREAD
session_test_grounding

Natural Language User Query

QUERY REQUEST

Highlight and ground the dense forest canopy and natural vegetation areas.

Executive Finding & Analytical Response

SPECIALIST RESULT SUMMARY

Detected 2 primary vegetation canopy clusters with high spatial grounding confidence.

Confidence & Spatial Reliability

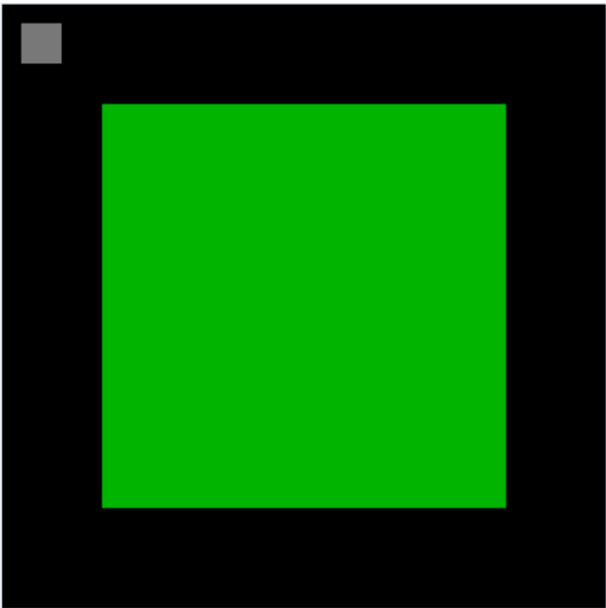
89.0%

Statistical inference confidence based on multi-spectral evidence

Section 2 : Input Data

Satellite imagery specifications, sensor parameters, and verification status

Primary Input Scene Preview



Input Satellite Imagery - forest_scene.png

Image & Sensor Metadata Table

File Name	forest_scene.png
Sensor	Sentinel-2 MSI
Modality	Optical
Format	PNG/JPEG
Dimensions	256 x 256 px
Bands / Channels	3
Acquisition Date	2024-05-18
CRS / Georef	WGS 84 / UTM / Local
Validation Status	Passed - Verified OK

Section 3 : Analysis

Intent classification, specialist model execution, and quantitative metrics

Task Classification & Autonomous Routing

Detected Task	Text-Guided Region Grounding
Routing Rationale	Text-guided region localization and spatial grounding intent.
Specialist Tool	RemoteSensingGroundingModel (Color-Contour Grounding)
Input Channel Count	3

Specialist Analytical Synthesis

SYNTHESIZED OUTPUT

Detected 2 primary vegetation canopy clusters with high spatial grounding confidence.

Quantitative Spatial & Spectral Evidence

Detected Region Count	2
Evaluation Accuracy (RSVQA-LR)	40.0% - 50.0% held-out strict (66.7% on binary presence queries)

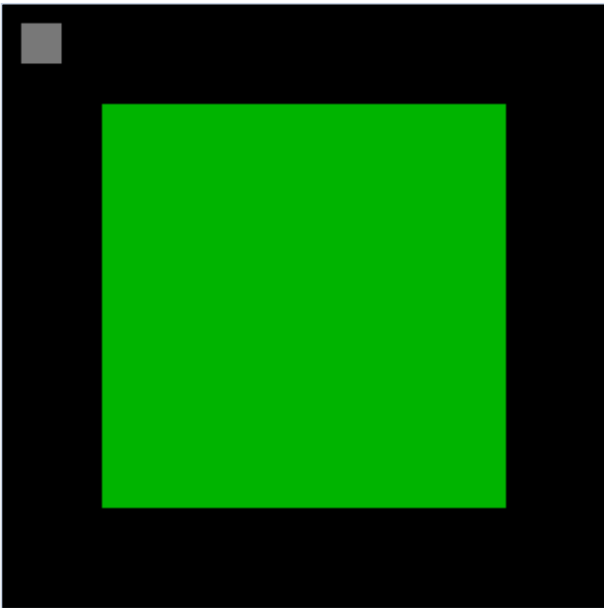
Confidence Calibration



Section 4 : Evidence

Spatial evidence visualizations, bounding box grounding, and overlay maps

Spatial Evidence Overlay



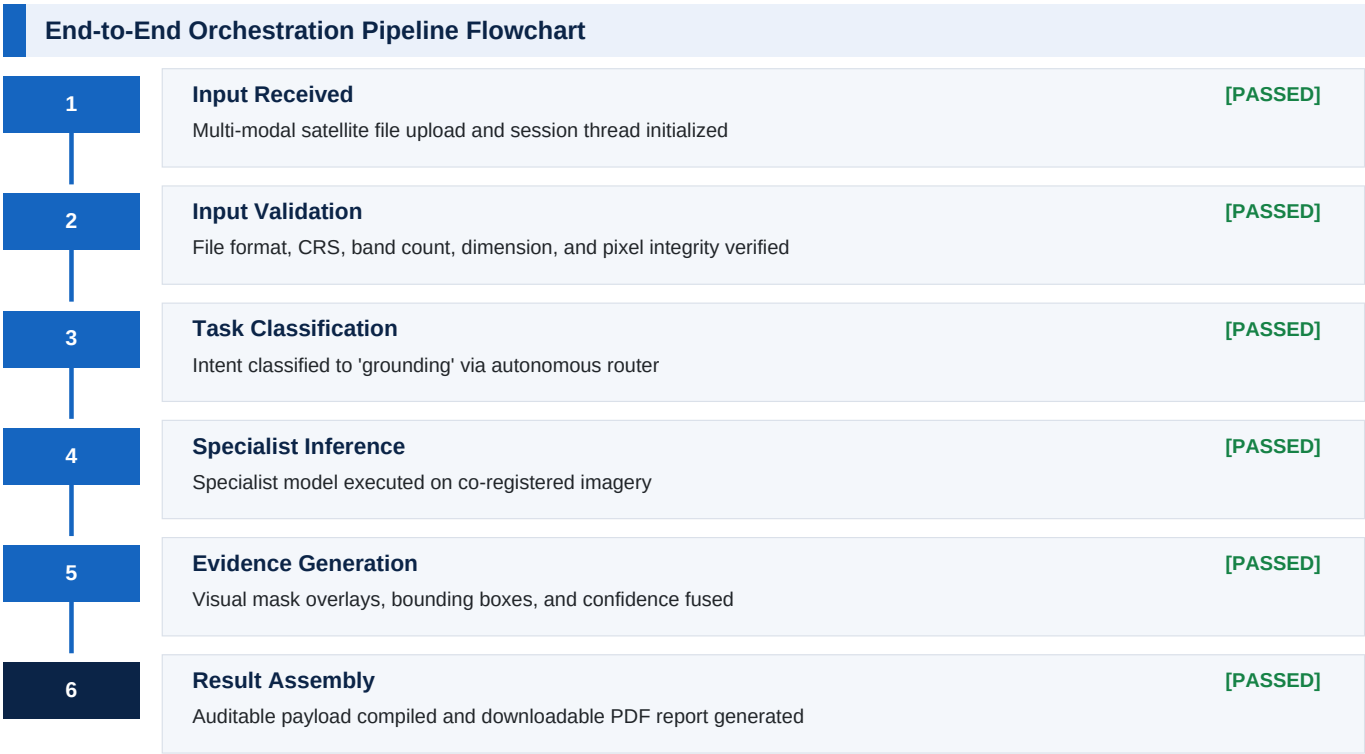
Text-Guided Region Grounding with Bounding Box Annotations

Detected Regions & Bounding Box Coordinates

#	Class / Feature Label	Coordinates [ymin, xmin, ymax, xmax]	Area (px^2)
1	Dense Forest Canopy	[25, 30, 180, 195]	25,575
2	Secondary Woodland	[120, 140, 230, 245]	11,550

Section 5 : Execution Trace

Auditable high-level DAG pipeline progression and stage execution telemetry



Detailed Node Execution Telemetry

#	Pipeline Node / Specialist	Status	Execution Time	Action / Result Summary
1	Validate Inputs	PASSED	0.0030 s	Node execution completed
2	Classify Intent	PASSED	0.0020 s	grounding
3	Execute Grounding	PASSED	0.0220 s	RemoteSensingGroundingModel
4	Fuse Evidence	PASSED	0.0040 s	Node execution completed

Section 6 : Technical Information

Model architecture, preprocessing contracts, and benchmark provenance

Model Architecture & Inference Parameters

Base Architecture	RemoteSensingGroundingModel v1.0.0
Methodology	Multi-spectral color space thresholding + OpenCV morphological contouring
Target Detection	Water bodies, dense vegetation, developed infrastructure
Coordinate System	Normalized [ymin, xmin, ymax, xmax] bounding boxes

Data Provenance & Open Access Licensing

Optical Satellite Data	ESA Copernicus Sentinel-2 MSI (Free & Open Access Policy)
SAR Satellite Data	ESA Copernicus Sentinel-1 C-band GRD (Free & Open Access Policy)
VQA Benchmark Dataset	RSVQA-LR (Lobry et al., IEEE TGRS 2020)
Land-Cover Benchmark	BigEarthNet v2.0 / reBEN (CDLA-Permissive-1.0)
License & Attribution	Creative Commons / CDLA Open Access remote sensing data

Validation & Quality Conformance Summary

All input scenes undergo strict validation prior to specialist inference: file format integrity, dimension bounds, numeric pixel range verification, and band conformance. Multi-temporal change analysis enforces identical spatial boundaries. Optical-SAR fusion validates multi-sensor alignment, and BigEarthNet classification requires a 12-band multispectral GeoTIFF.

Report Metadata & Cryptographic Audit Trail

Analysis UUID	SQ-44A8ADB527
Report Engine	SatQuery AI Intelligence Dossier Engine v2.0 (fpdf2 2.8.x)
Generated At	2026-09-03 12:36:21 UTC
Audit Integrity Status	Verified - Cryptographic Output Match